

Advanced Situated Visualizations for Sports Videos

PhD Position

Project: ANR-funded SportsViz (2025-2029) <https://sportsviz.projet.liris.cnrs.fr/>

Expected dates: As soon as possible (eg Dec. 2025) possibility of Internship prior do the PhD

Advisors: Romain Vuillemot (PI, Centrale Lyon) <https://romain.vuillemot.net/>

Co-advisors: Petra Isenberg (Inria Saclay), Lionel Reveret (Inria Grenoble)

Keywords: sports data visualization, situated visualizations, computer graphics.

Gross monthly salary: around € 2,100 – € 2,200 (+ extra funding if teaching)

Context: Data visualization is a research field that aims to empower humans with decision-making and communication capabilities through graphical displays. While data visualizations have strong narrative and analytical power for sports analysis [8], they are currently not integrated where and when they are most relevant—in the space where data are usually collected: videos. The standard way scientists or broadcasters display data is by using visualization inserts superimposed on the video rather than situating them within the sports 3D scene, e. g., on a wall, thereby leveraging the physical context to show progress, emphasize key performance factors (e. g., forces, speed), and put them into perspective with competitors. The resulting situated visualization would enable an informed decision-making process for strategic choices and also engage a broader audience, such as sports fans [7], while preserving and enhancing the viewing experience. **This PhD position is dedicated to improving athletes performance and audience engagement through advanced situated visualizations [2] in sports videos.** This proposal is part of SportsViz, an ANR-funded (2025–2029) research project dedicated to improving athletes’ performance and audience engagement with such situated videos.

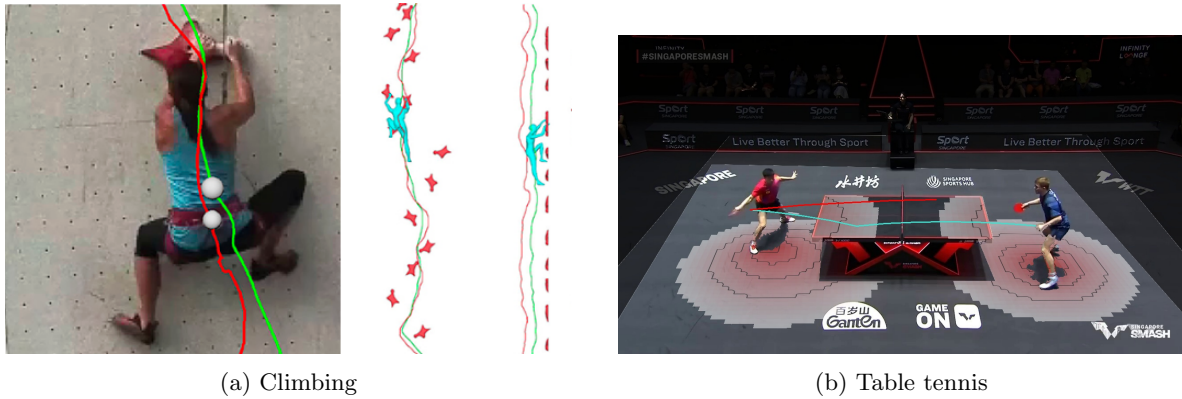


Figure 1: Example of situated sports visualizations as inspiration for this PhD. The left image shows speed climbing with an overlay of a 3D trajectory captured from video [9]. The right shows a static overlay of shot probabilities in table tennis [4]. Both suffer from disadvantages like visual clutter, lack of integration in the sports scene and distortion due to camera perspective.

Challenge: The main challenge in designing situated sports visualization is to capture and communicate the sports narrative, i. e. the key sports events that form the game scenario, from beginning to finish. It is crucial to have additional knowledge beyond positions, events and actions data of the game. This knowledge is typically acquired with a mixture of game rules and performance analysis that camera operators and directors take into account to select camera angles and shots, supported by sports commentators. We should learn from these approaches and capture them as data-driven rules so the right data and visualization will be mapped at the right place at the right time.

Goal of the PhD: The goal of this PhD is to develop novel situated visualizations for sports videos. These visualizations must address a range of constraints, including irregular mapping shapes, dynamic scenes, and changing points of view. They should support the needs of both experts and general audiences engaged in sports analysis. To tackle these challenges, the candidate will build an infrastructure for rapidly prototyping situated visualizations that combine visualization, interaction techniques, and complex 3D displays, applied to sports data and scene we collected in the project. Such infrastructure should facilitate the ideation process through examples and templates [11, 3], enabling the reuse of pre-built 3D scenes and objects while guiding designs toward best practices. This milestone aims to establish a large, structured collection of examples and design guidelines as starting points for the design process, identify reusable patterns across different sports; and foster creative design by providing expert users (i. e. designers) with a high degree of flexibility.

The PhD candidate will be assisted by a research engineer who will provide the video infrastructure, tracking data and video broadcast platform. Interns with design training will be hired to provide feedback and improve the tool, and design visualizations to reach two distinct communities: sports fans and TV broadcasters on one side (general audience), coaches, athletes and sport analysts on the other (expert audience). We expect novel visualizations to emerge, as well as novel communication narratives [10]. The PhD candidate will also be assisted by other partners providing 3D models of the scenes and tracking data.

Organization of the PhD: We plan to organize the PhD as follows:

1. **State of the art** that reviews similar authoring tools for 3D graphics, data visualization, e. g., [6, 1, 5] or commercial video editors. Identify design patterns, existing interaction techniques common to all those domains.
2. **Authoring infrastructure design** that combines (existing) 3D scene reconstruction with 2D assets such as visualization, tracking data, and propose interactions that enable to quickly explore situated visualization design.
3. **Implementation of the authoring tool and validation** with focus group and deployment to sports partners and fans. It is expected to improve the tool based on user feedback and novel use case (e. g., new sports to be supported).
4. **Publications:** IEEE Vis, ACM CHI, IUI conferences and/or journals such as TVCG, ToCHI, and specialized conferences or workshops on sports (e. g., MLSA).

The following skills are expected for the candidate:

1. Very good programming skills, with knowledge in 3D graphics and/or 2D visualization design
2. Proficient in English (written and spoken)
3. Interested in sport data analysis and visualization
4. Willing to pursue a career in Research or Research and Development
5. Familiar or willing to learn computer vision and deep learning techniques related to motion and pose estimation.

Partners: The PhD will be part of the SportsViz project that involves Ecole Centrale de Lyon, Inria Saclay and Inria Grenoble. **Romain Vuillemot (Coordinator)** - <https://romain.vuillemot.net/> is an assistant professor (Maître de Conférence) at Ecole Centrale de Lyon and LIRIS Lab (UMR 5205) in the SICAL team. His expertise is in Information Visualization and Visual Analytics, with a particular focus on sports analytics. He is the coordinator at Ecole Centrale de Lyon of the *Sciences*²⁰²⁴ network. He previously lead the video tracking package for the PIA NePTUNE (2020 - 2024) dedicated to swimmers tracking in competition. He is leading the research on visual analysis of tactics in table tennis <https://centralelyon.github.io/table-tennis/>.

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